Exp 23 MATLAB CODE

```
clc;
clear;
close all;

% Parameters
M = 16;
N = 4000;
SNR = 15;
k = log2(M);
sps = 8;

% Generate Random Data
data = randi([0 1],N*k,1);

% Convert Bits to Symbols
data_matrix = reshape(data,k,[])';
weights = 2.^(k-1:-1:0)';
symbol_indices = data_matrix * weights;

% M-QAM Modulation
tx = qammod(symbol_indices,M,'gray','UnitAveragePower',true);

% Add AWGN Noise
rx = awgn(tx,SNR,'measured');

% M-QAM Demodulation
rx_indices = qamdemod(rx,M,'gray','UnitAveragePower',true);
```

```
% Convert Symbols to Bits
detected_matrix = de2bi(rx_indices,k,'left-msb');
detected_data = reshape(detected_matrix',[],1);

% BER Calculation
errors = sum(data ~= detected_data);
BER = errors/length(data);

fprintf('M-QAM Order = %d\n',M);
fprintf('SNR = %d dB\n',SNR);
fprintf('Total Bits = %d\n',length(data));
fprintf('Bit Errors = %d\n',errors);
fprintf('Bit Error Rate = %.6f\n',BER);

% Constellation Diagram
figure;
subplot(1,2,1);
plot(real(tx),imag(tx),'b.');
title('Transmitted M-QAM Constellation');
xlabel('In-Phase');
ylabel('Quadrature');
grid on;
axis equal;

subplot(1,2,2);
plot(real(rx),imag(rx),'r.');
title('Received M-QAM Constellation with AWGN');
```

```
xlabel('In-Phase');
ylabel('Quadrature');
grid on;
axis equal;

% Eye Diagram using Oversampled QAM Symbols
tx_wave = repelem(tx(1:100),sps);
rx_wave = repelem(rx(1:100),sps);

figure;
subplot(2,1,1);
eyediagram(real(tx_wave),2*sps);
title('Transmitted M-QAM Eye Diagram');

subplot(2,1,2);
eyediagram(real(rx_wave),2*sps);
title('Received M-QAM Eye Diagram with AWGN');
```

